AAE 521 Plasma Lab – Spring 2026
3 credits

1 Course Description

The laboratory course will include lab prep lectures, practical classes and mini-projects. Lab prep lectures will briefly cover topics of basic plasma physics and diagnostics relevant to the subsequent practical classes. In addition, details of the corresponding lab procedure, instructions and lab report assignment will be discussed. Practical classes will involve students in practical creation and operation of various plasma sources and plasma diagnostics. Specifically, students will operate DC high voltage breakdown facility, electrostatic accelerator (ion thruster), cross-field accelerator (Hall thruster), Atmospheric-Pressure Plasma Jet facility, and will conduct measurements of plasma parameters using Langmuir probes, microwave interferometer and optical spectrometer.

2 Learning Resources and Reference Text

Brightspace Learning Management System: All course materials will be posted on the course Brightspace site.

AI Policy: You are allowed to use AI tools in this course.

Textbook: I.H. Hutchinson, Principles of plasma diagnostics, New York: Cambridge University Press, 2nd ed., 2002.

3 Course Learning Objectives

By the end of the semester, successful students will be able to:

- To operate various plasma sources including electrostatic accelerator, Hall accelerator, glow discharge and atmospheric pressure discharge.
- To demonstrate practical skills in conducting measurements of basic electrical discharge parameters and thrust.
- To demonstrate practical skills in conducting measurements of plasma parameters using electrostatic Langmuir probes, optical spectroscopy and microwave interferometry.
- To demonstrate fundamental understanding of basic concepts utilized in studied plasma diagnostic techniques.
- To demonstrate fundamental understanding of various breakdown mechanisms including Townsend and streamer breakdown.

4 Assignments

4.1 Pre-Lab assignment

Pre-lab assignments have to be submitted in Brightspace prior to the lab session.

4.2 Lab Reports

Lab Reports for each lab have to be submitted in Brightspace.

4.3 Mini Projects

Mini Reports for each lab have to be submitted in Brightspace.

4.4 CATME Peer Feedback

The scope of the CATME Peer Feedback includes the quality and quantity of work evaluated by your peer students. The CATME Peer Feedback grade is relative to your subteam and the rest of the class and will be determined by feedback both from subteam members.

5 Evaluation

Course grading:	Pre-Lab Assignment	= 5%
	Lab Reports	= 60%
	Mini Projects	= 35%
	TOTAL	= 100%

Grading schema: A+ (≥97), A (≥93), A- (≥90), B+(≥87), B (≥83), B- (≥80), C+ (≥77), C (≥73), C- (≥65), D (≥60), F (<60)